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SHREE L. R. TIWARI COLLEGE OF ENGINEERING

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DEPARTMENT OF COMPUTER ENGINEERING

CML610 Cloud Computing Lab

Sixth Semester, 2025-2026 (Even Semester)

Name of Student : Mainuddin Nayeem Khan

Roll No. : 53

Division : A

Batch : A2

Day / Session : Monday/Friday

Venue : 310

Experiment No. :4

Title of Experiment : Study of Infrastructure as a Service

Date of Conduction :

Date of Submission :

Particulars	Max. Marks	Marks Obtained
Preparedness and Efforts(PE)	**3**	
Knowledge of tools(KT)	**3**	
Debugging and results(DR)	**3**	
Documentation(DN)	**3**	
Punctuality & Lab Ethics(PL)	**3**	
Total	**15**	

Grades – Meet Expectations (3 Marks), Moderate Expectations (2 Marks), Below Expectations (1 Mark)

Checked and Verified by

Name of Faculty : Pravin Jangid

Signature :

Date :

Experiment 4

Aim:

Study Infrastructure as a Service

Theory:

1. Detailed Study of Infrastructure as a Service (IaaS)

Infrastructure as a Service (IaaS) is one of the core service models of cloud computing that provides **virtualized computing resources over the internet**. In this model, users rent IT infrastructure such as **servers, storage, networking components, and virtualization technology** from a cloud service provider instead of purchasing and maintaining physical hardware.

In traditional IT infrastructure, organizations must invest heavily in servers, networking devices, power supply, cooling systems, physical space, and maintenance staff. IaaS eliminates this burden by allowing users to access infrastructure resources on demand through a cloud platform.

How IaaS Works

- IaaS operates on the principle of virtualization. Physical servers in data centers are divided into multiple virtual machines using hypervisors. Each virtual machine acts like an independent computer with its own operating system, applications, and resources.
- The cloud service provider manages:
- Physical servers and hardware
- Data centers
- Power supply and cooling

- Network infrastructure
- Virtualization layer (hypervisor)
- The user manages:
- Operating systems
- Installed software and applications
- Middleware and runtime
- Data and security configurations

Key Components of IaaS

Compute Resources
These include virtual machines (VMs) that provide processing power. Users can choose the number of CPUs, amount of RAM, and type of operating system.

1. **Storage Services**

IaaS provides different types of storage:

Block storage for operating systems and databases

Object storage for files, backups, and multimedia

Archive storage for long-term data retention

2. **Networking Services**
   Includes virtual networks, subnets, IP addresses, gateways, load balancers, and firewalls that help manage communication between resources.
3. **Virtualization Technology:** Hypervisors such as VMware, KVM, or Hyper-V are used to create and manage virtual machines.
4. **Management Interfaces :** Web-based dashboards and APIs allow users to configure, monitor, and manage their infrastructure remotely.

Use Cases of IaaS

- Hosting web and enterprise applications
- Software development and testing
- Backup and disaster recovery solutions

- Big data analytics and machine learning
- Running legacy applications
- High-performance computing workloads

Examples of IaaS Providers

- Amazon Web Services (EC2)
- Microsoft Azure Virtual Machines
- Google Compute Engine
- IBM Cloud Infrastructure
- Oracle Cloud Infrastructure

2. Advantages and Limitations of Infrastructure as a Service (IaaS)

Advantages of IaaS

1. Reduced Capital Investment :Organizations do not need to buy expensive servers, networking devices, or data center infrastructure. This significantly reduces upfront costs and allows businesses to invest resources elsewhere.

2. Scalability and Elasticity :IaaS allows users to scale resources up or down instantly based on demand. This is ideal for applications with variable workloads such as e-commerce websites or seasonal services.

3. High Flexibility and Customization: Users can select their preferred operating systems, software, development tools, and configurations. This level of control is not possible in higher cloud models like SaaS.

4. Faster Time to Market : Infrastructure provisioning that once took weeks or months can now be completed within minutes, enabling faster application development and deployment.

5. Global Accessibility : IaaS services can be accessed from anywhere in the world through the internet, supporting remote work and global business operations.

6. Reliability and Availability : Most cloud providers offer redundancy, automated backups, and disaster recovery mechanisms to ensure high availability of services.

7. Pay-as-You-Go Pricing : Users are charged only for the resources they consume, making cost management more efficient if resources are used wisely.

Limitations of IaaS

1. Security Management Responsibility: While the provider secures the physical infrastructure, users are responsible for securing operating systems, applications, and data, which increases risk if mismanaged.

2. Requires Skilled Professionals: Managing virtual machines, networks, and security configurations requires technical expertise, which may not be available in small organizations.

3. Risk of Cost Overruns: Improper monitoring or unused resources can result in high and unexpected cloud bills.

4. Dependency on Internet Connectivity: Access to IaaS resources depends completely on a stable and high-speed internet connection.

5. Vendor Lock-In: Switching from one cloud provider to another can be complex due to differences in platforms, tools, and services.

6. Performance Variability : Since infrastructure is shared among multiple users, performance may vary due to resource contention.

7. Downtime and Service Outages : Although rare, outages at the provider's data center can disrupt services and affect business operations.

3. Security Issues in Infrastructure as a Service (IaaS)

Security is one of the most critical concerns in IaaS due to its shared and internet-based nature. IaaS follows a **shared responsibility model**, where both the cloud provider and the user play roles in maintaining security.

Major Security Issues in IaaS

1. Data Breaches:Sensitive data stored in cloud systems can be exposed due to weak authentication, lack of encryption, or unauthorized access.

2. Insecure APIs and Interfaces :Cloud services are controlled through APIs and web interfaces. Poorly designed or unsecured APIs can be exploited by attackers.

3. Virtual Machine Attacks : Vulnerabilities in the hypervisor or virtual machine isolation can allow attackers to access other users' data or systems.

4. Account Hijacking :Phishing attacks, stolen credentials, or weak passwords can allow attackers to gain control of cloud accounts.

5. Misconfiguration Errors

Improper security settings such as open ports, public storage access, or weak firewall rules are common causes of cloud security incidents.

6. Denial of Service (DoS and DDoS) Attacks

Attackers may overload cloud resources, making applications unavailable to legitimate users.

7. Data Loss: Accidental deletion, ransomware attacks, or lack of proper backup strategies can lead to permanent data loss.

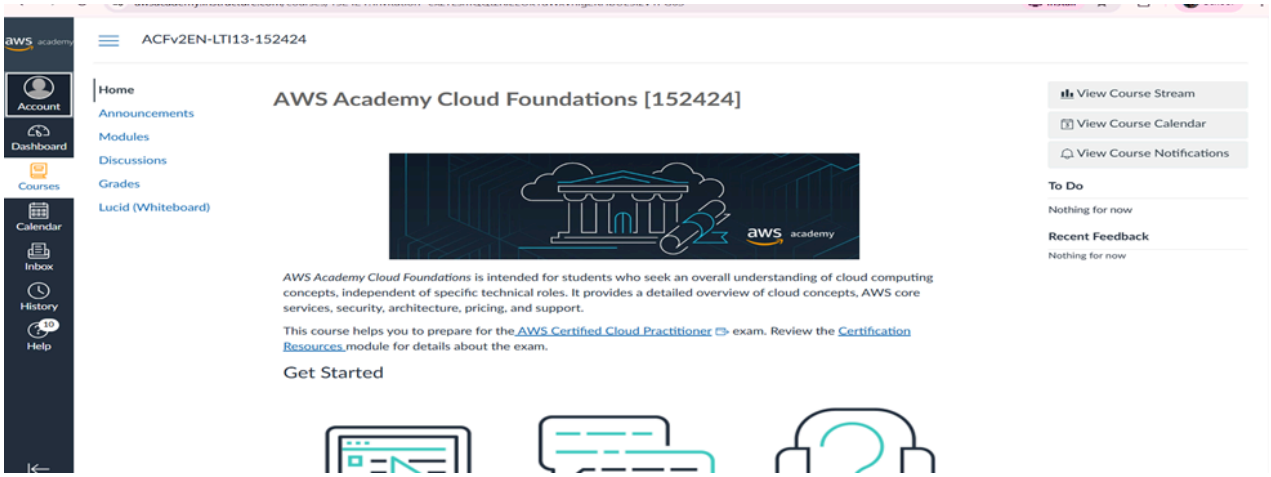
8. Insider Threats :Malicious or careless employees of either the user organization or the cloud provider can compromise security.

Activity

1. Creating and Configuring a VM using AWS EC2

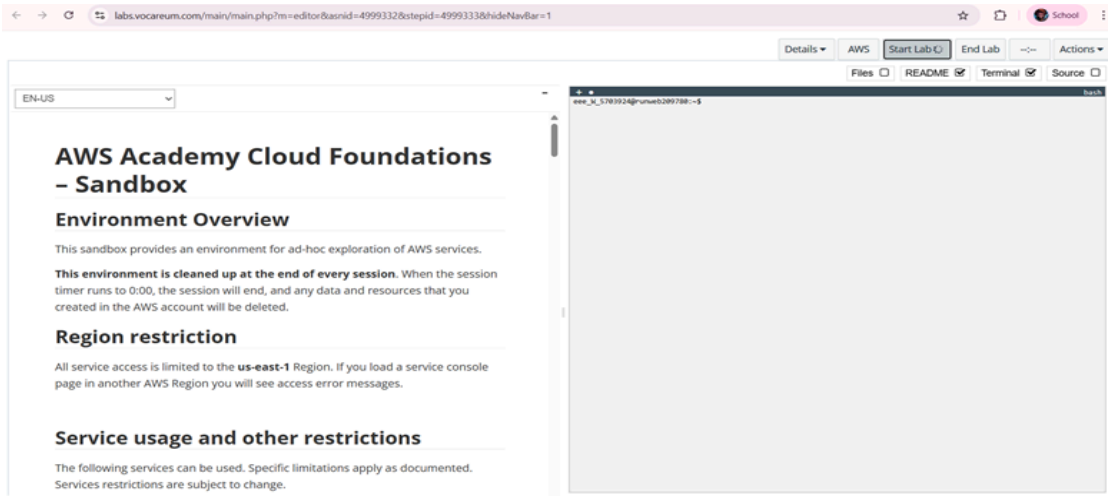
Step 1: Login to AWS Console

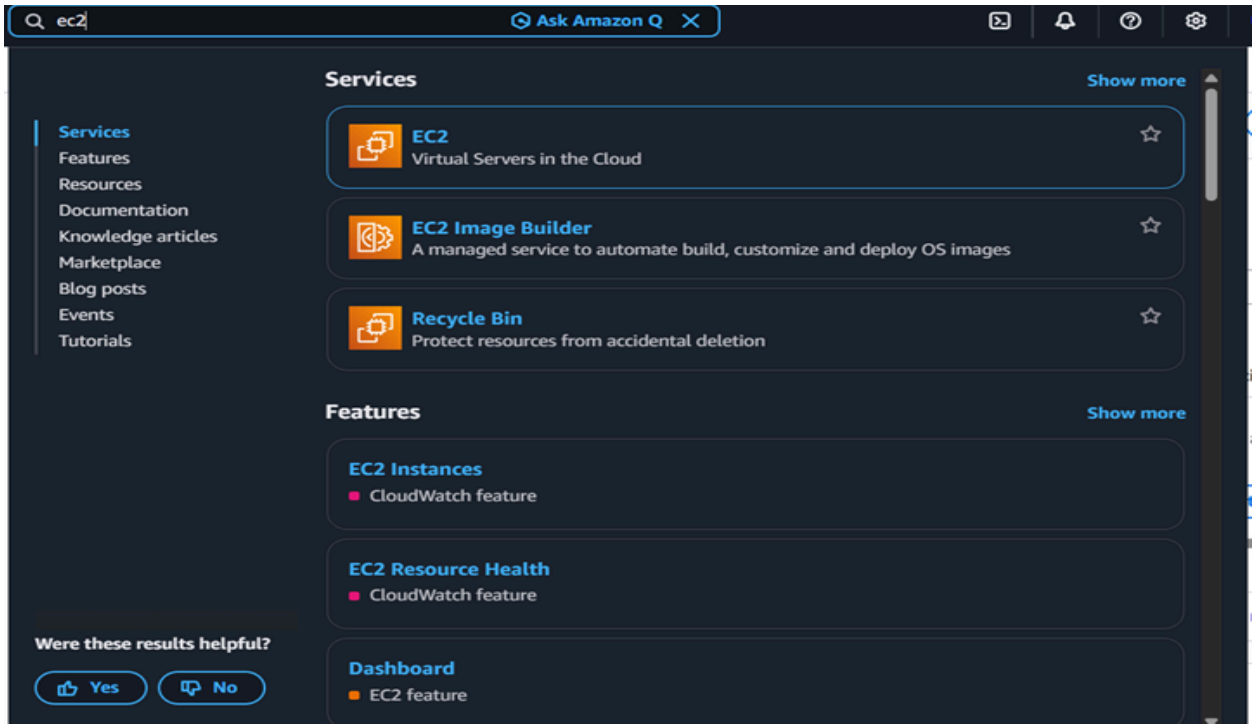
1. Open browser → go to **AWS Management Console**
2. Login using AWS credentials
3. Select a nearby **Region** (e.g., Mumbai – ap-south-1)



Step 2: Launch EC2 Instance

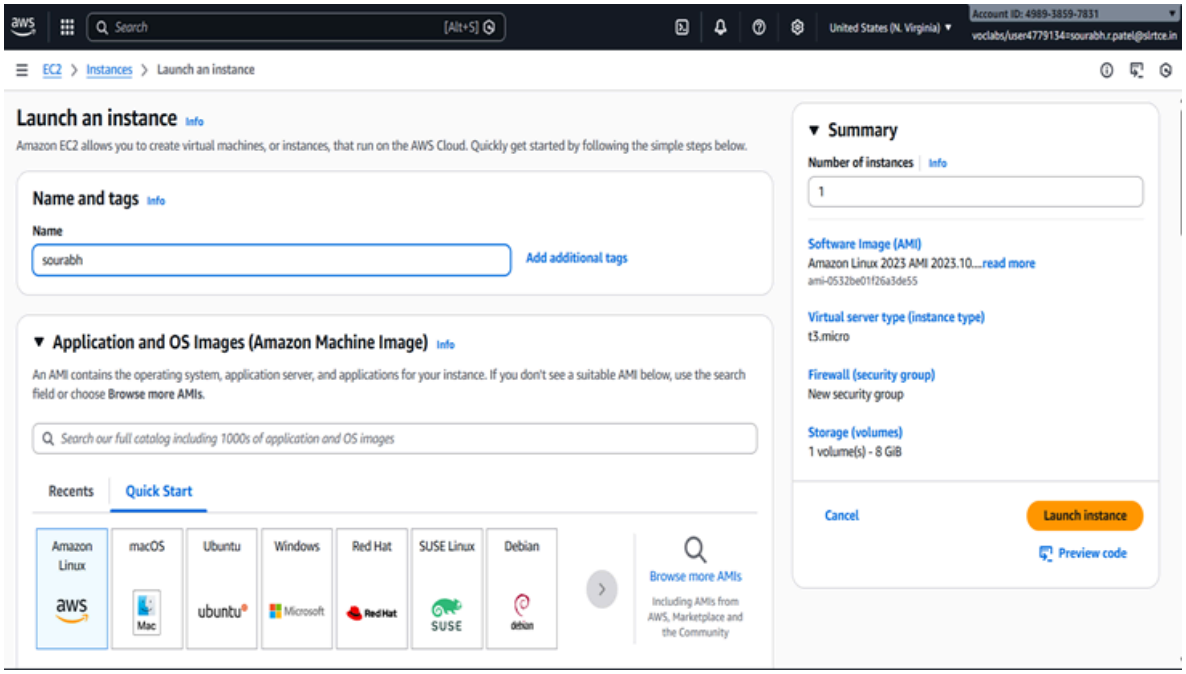
1. Go to **Services** → **EC2**
2. Click **Launch Instance**
3. Enter instance name (e.g., `MyEC2VM`)





Step 3: Choose AMI (Operating System)

- **Linux:** Amazon Linux 2 / Ubuntu
- **Windows:** Microsoft Windows Server

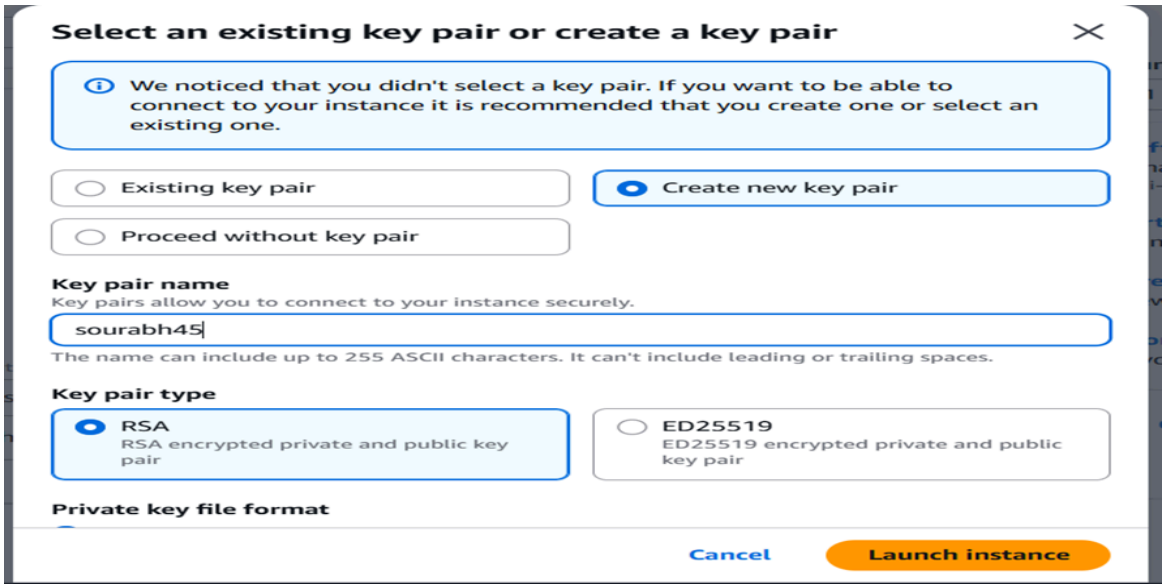


Step 4: Choose Instance Type

- Select **t2.micro (Free Tier eligible)**

Step 5: Create Key Pair

1. Click **Create new key pair**
2. Key pair type: RSA
3. Download the `.pem` file (important for login)



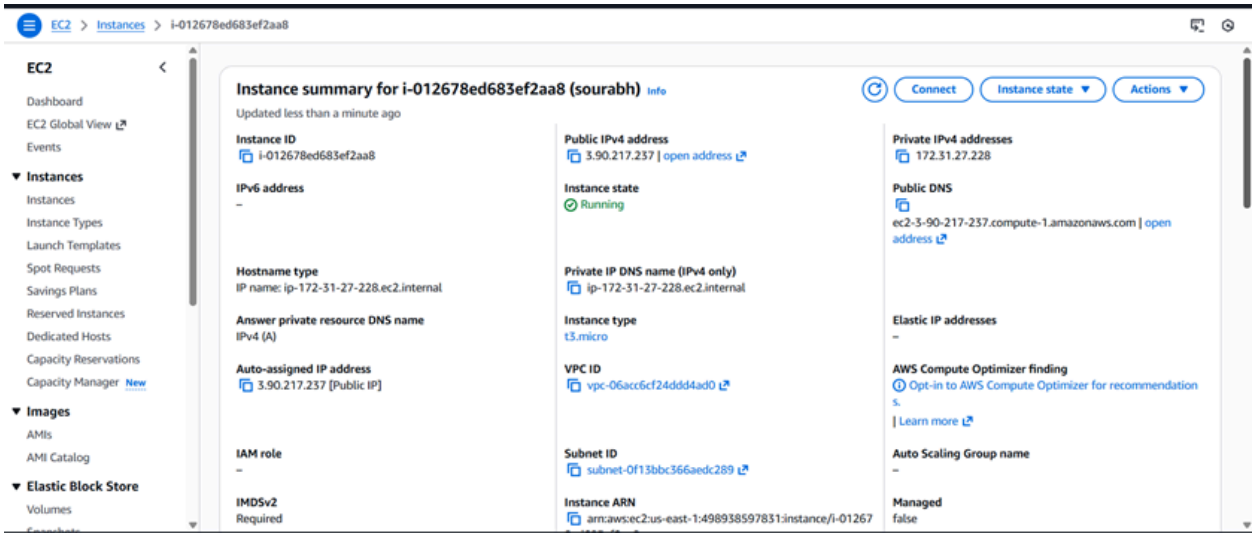
Step 6: Configure Security Group

Allow required inbound rules:

- **Linux**
  - SSH → Port **22**
- **Windows**
  - RDP → Port **3389**
- Source: `0.0.0.0/0`

Step 7: Launch Instance

Click **Launch Instance**
Instance state will change from **Pending** → **Running**



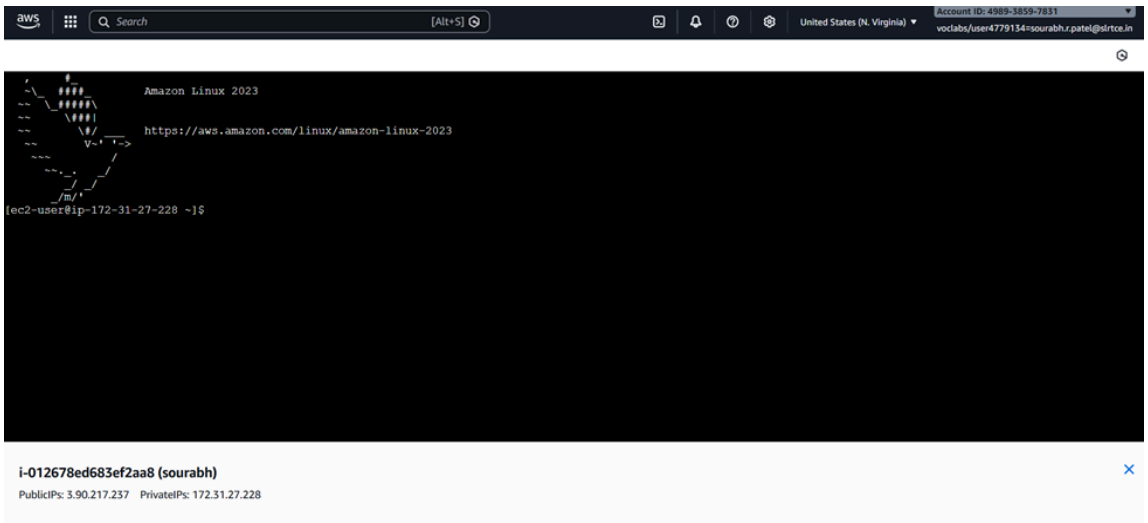
2. Run Linux/Windows Virtual machine inside Amazon EC2

Steps to Run Linux VM inside Linux EC2

Step 1: Launch EC2

- AMI: **Amazon Linux 2**

- Instance Type: **t3.large**
- Storage: **30 GB** ● Security Group



: Allow SSH (22)

Step 2: Check Virtualization Support

```
egrep -c '(vmx|svm)' /proc/cpuinfo
```

Output **> 0** → Virtualization supported ✅

Step 3: Install KVM Hypervisor

```
sudo yum update -y
```

```
sudo yum install qemu-kvm libvirt virt-install bridge-utils -y
```

Start services:

```
sudo systemctl start libvirtd
```

```
sudo systemctl enable libvirtd
```

Now you can create **Linux virtual machines** inside EC2.

Windows VM inside EC2

- Use **Windows EC2**
- Enable **Hyper-V**
- Create VM inside Windows
  ⚠️ Performance is low, **not recommended for production**.

2 Installation of Java in Amazon Linux

Now we work **directly on Amazon Linux EC2**.

Step 1: Connect to EC2

```
ssh -i key.pem ec2-user@<EC2-Public-IP>
```

Step 2: Install Java (OpenJDK)

Amazon Linux supports **OpenJDK**.

Install Java 11

```
sudo yum install java-11-openjdk -y
```

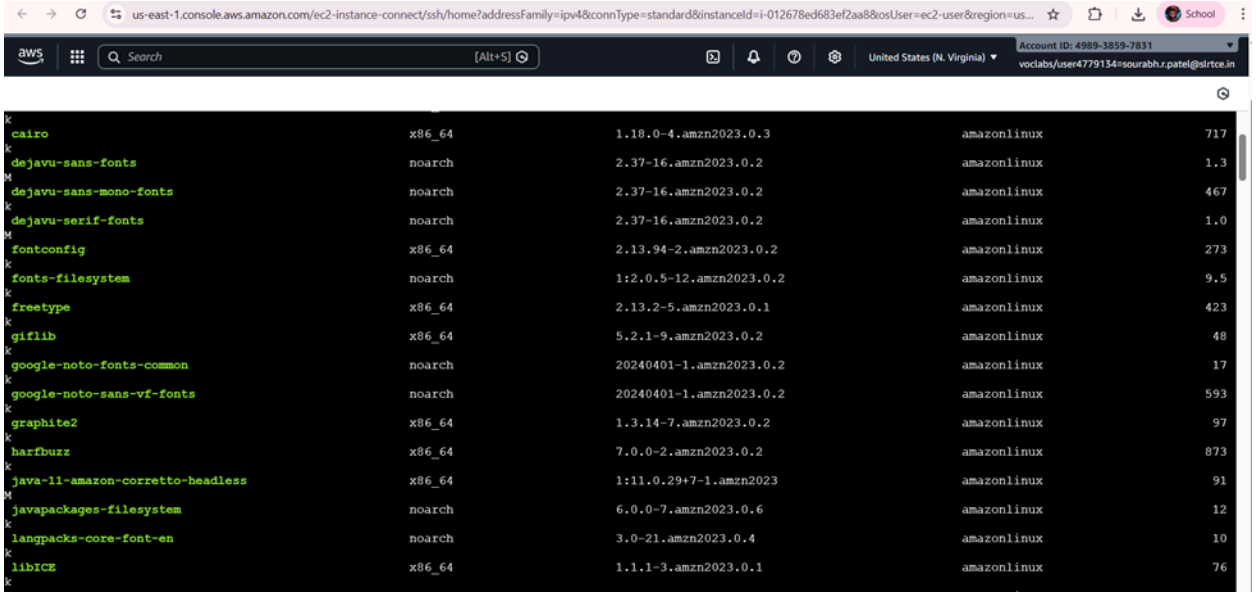
Or Java 8:

```
sudo yum install java-1.8.0-openjdk -y
```

Step 3: Verify Java Installation

```
java -version
```

Example Output:




```
openjdk version "11.0.x"
```

3 Java Hello World Program in Amazon Linux

Step 1: Create Java File

```
nano HelloWorld.java
```

Step 2: Write Java Code

```
class HelloWorld {
      public static void main(String[] args) {
      System.out.println("Hello World from Amazon Linux EC2");
      }
}
```

Save and exit:

- CTRL + X → Y → Enter

Step 3: Compile Java Program

```
javac HelloWorld.java
```

Step 4: Run Java Program

```
java HelloWorld
```

```
Caused by: java.lang.ClassNotFoundException: Helloworld
[ec2-user@ip-172-31-27-228 ~]$ ^C
[ec2-user@ip-172-31-27-228 ~]$ ls
Helloworld.java
[ec2-user@ip-172-31-27-228 ~]$ javac Helloworld.java
-bash: javac: command not found
[ec2-user@ip-172-31-27-228 ~]$ java -version
openjdk version "11.0.29" 2025-10-21 LTS
OpenJDK Runtime Environment Corretto-11.0.29.7.1 (build 11.0.29+7-LTS)
OpenJDK 64-Bit Server VM Corretto-11.0.29.7.1 (build 11.0.29+7-LTS, mixed mode)
[ec2-user@ip-172-31-27-228 ~]$ i
-bash: i: command not found
[ec2-user@ip-172-31-27-228 ~]$ ^C
[ec2-user@ip-172-31-27-228 ~]$ java Helloworld
Error: Could not find or load main class Helloworld
Caused by: java.lang.ClassNotFoundException: Helloworld
[ec2-user@ip-172-31-27-228 ~]$ nano Hello.java
[ec2-user@ip-172-31-27-228 ~]$ javac Hello.java
-bash: javac: command not found
[ec2-user@ip-172-31-27-228 ~]$ sudo yum install java-11-openjdk-devel -y
Last metadata expiration check: 0:22:06 ago on Sun Feb  8 01:37:03 2026.
No match for argument: java-11-openjdk-devel
Error: Unable to find a match: java-11-openjdk-devel
[ec2-user@ip-172-31-27-228 ~]$ sudo yum install java-11-openjdk-devel-y
Last metadata expiration check: 0:22:43 ago on Sun Feb  8 01:37:03 2026.
No match for argument: java-11-openjdk-devel-y
Error: Unable to find a match: java-11-openjdk-devel-y
[ec2-user@ip-172-31-27-228 ~]$ sudo dnf install java-11-openjdk-devel -y
Last metadata expiration check: 0:23:50 ago on Sun Feb  8 01:37:03 2026.
No match for argument: java-11-openjdk-devel
Error: Unable to find a match: java-11-openjdk-devel
[ec2-user@ip-172-31-27-228 ~]$ cat > HelloWorld.java
[ec2-user@ip-172-31-27-228 ~]$ javac HelloWorld.java
[ec2-user@ip-172-31-27-228 ~]$ java HelloWorld
Hello World
[ec2-user@ip-172-31-27-228 ~]$
```

3. Access the created machine remotely

Step 1: Configure Port Forwarding on EC2

Example:

- EC2 port **2222** → VM port **22**

Step 2: Add Inbound Rule

Security Group:

- Custom TCP
- Port: **2222**
- Source: `0.0.0.0/0`

Step 3: PuTTY Configuration

Host Name:

```
ec2-user@<EC2-Public-IP>
```

In PuTTY:

Connection → SSH → Tunnels

- Source port: `2222`
- Destination: `192.168.122.10:22`
- Click **Add**

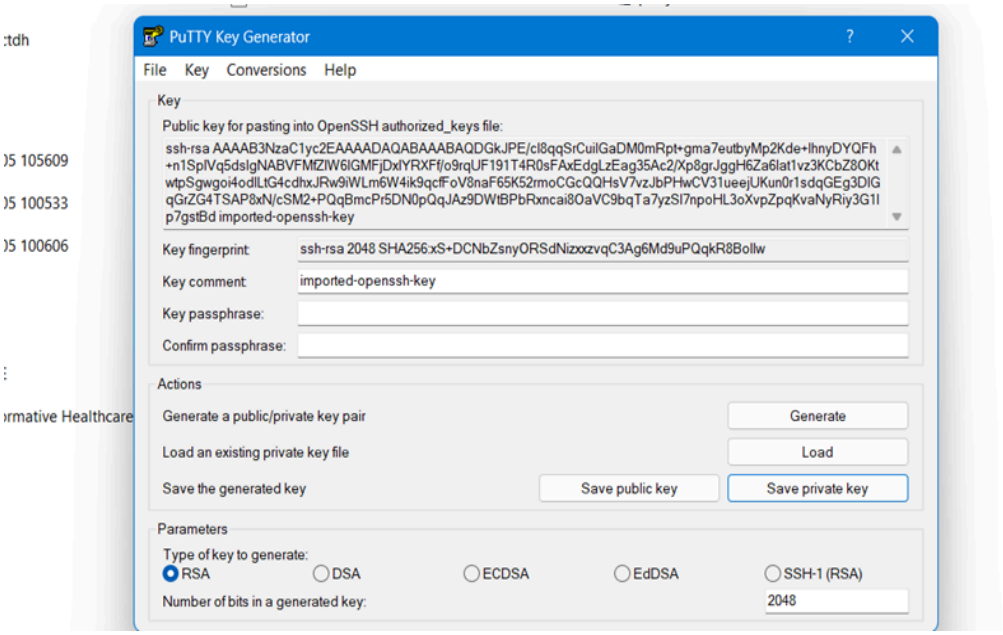
Attach `.ppk` key → **Open**

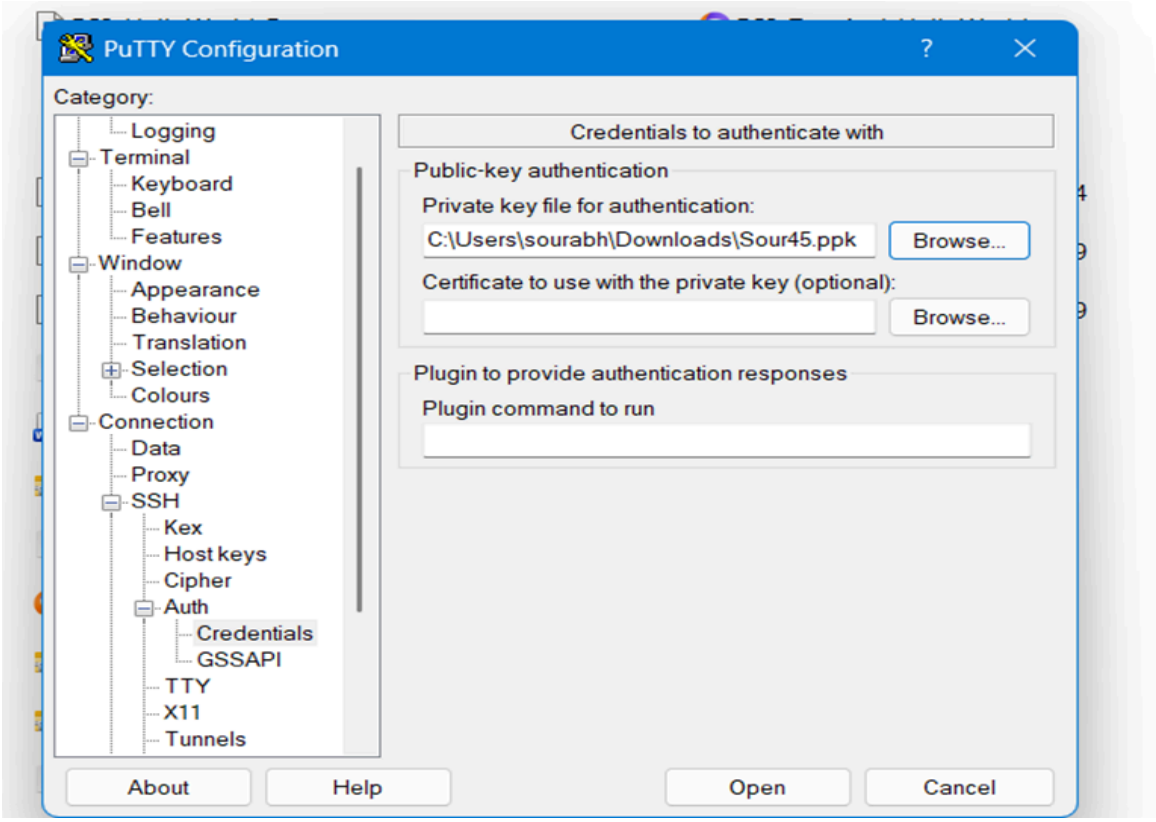
Step 4: SSH into VM

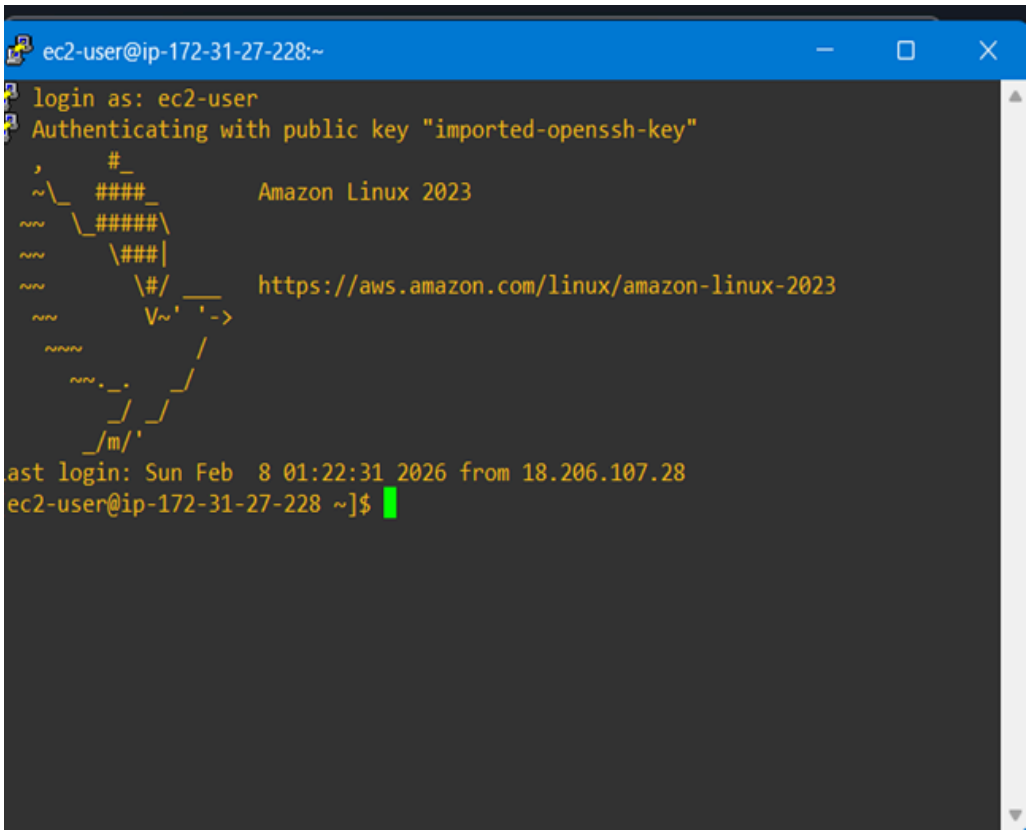
From PuTTY terminal:

```
ssh ubuntu@localhost -p 2222
```

✅ Virtual machine accessed **remotely using PuTTY**.


PuTTY Key Generator
File Key Conversions Help
Key
Public key for pasting into OpenSSH authorized_keys file:
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQDGkJPE/cl8qqSrCuilGaDM0mRpt+gma7eutbyMp2Kde+lhnyDYQFh
+n1SpIVq5dsIgNABVFMfZIW6IGMFjDxIYRXFf/o9rqUF191T4R0sFAxEdgLzEag35Ac2/Xp8grJggH6Za6lat1vz3KCbZ8OKt
wtpSgwgoi4odlLtG4cdhxJRw9iWLm6W4ik9qcfFoV8naF65K52rmoCGcQQHsV7vzJbPHwCV31ueejUKun0r1sdqGEg3DlG
qGrZG4TSAP8xN/cSM2+PQqBmcPr5DN0pQqJAz9DWtBPbRxncai8OaVC9bqTa7yzSI7npoHL3oXvpZpqKvaNyRiy3G1l
p7gstBd imported-openssh-key
Key fingerprint: ssh-rsa 2048 SHA256:xS+DCNbZsnyORSdNizxxzvqC3Ag6Md9uPQqkR8Bollw
Key comment: imported-openssh-key
Key passphrase:
Confirm passphrase:
Actions
Generate a public/private key pair
Generate
Load an existing private key file
Load
Save the generated key
Save public key
Save private key
Parameters
Type of key to generate:
RSA
DSA
ECDSA
EdDSA
SSH-1 (RSA)
Number of bits in a generated key:
2048


PuTTY Configuration
Category:
Logging
Terminal
Keyboard
Bell
Features
Window
Appearance
Behaviour
Translation
Selection
Colours
Connection
Data
Proxy
SSH
Kex
Host keys
Cipher
Auth
Credentials
GSSAPI
TTY
X11
Tunnels
Credentials to authenticate with
Public-key authentication
Private key file for authentication:
C:\Users\sourabh\Downloads\Sour45.ppk
Browse...
Certificate to use with the private key (optional):
Browse...
Plugin to provide authentication responses
Plugin command to run
About
Help
Open
Cancel



Conclusion

Benefits of using IaaS (Infrastructure as a Service)

1. **Cost Effective**
   No need to buy physical servers or hardware. You pay only for what you use, which reduces upfront and maintenance costs.
2. **Scalability and Flexibility**
   Resources like CPU, storage, and RAM can be increased or decreased anytime based on demand.
3. **High Availability and Reliability**
   Cloud providers offer backups, disaster recovery, and multiple data centers, ensuring systems stay available.
4. **Faster Deployment**
   Servers and virtual machines can be created in minutes instead of weeks, speeding up application development.
5. **Better Performance**
   IaaS providers use modern hardware and optimized infrastructure, giving better performance.

6. **Remote Accessibility**
   Infrastructure can be accessed and managed from anywhere using the internet.
7. **Security Management by Provider**
   Physical security, data center security, and hardware protection are handled by the cloud provider.
8. **Easy Maintenance and Updates**
   No need to worry about hardware upgrades or failures; the provider manages them.